Quiz 1 Solutions, MA2031, Aug 28, 2023

1. Let $A = \begin{bmatrix} 1 & 1 & 3 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 3 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 6 \\ 4 \\ 3 \\ 2 \\ 6 \end{bmatrix}$.

   Determine $\mathrm{Sol}(A, b)$ by using Gauss-Jordan elimination. [6]

   Solution: $[A|b] \xrightarrow{RREF} \left[\begin{array}{cccc|c} \boxed{1} & 0 & 1 & 0 & 2 \\ 0 & \boxed{1} & 2 & 0 & 1 \\ 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$. [**4**]

   $\xrightarrow{del-0\ rows} \left[\begin{array}{cccc|c} \boxed{1} & 0 & 1 & 0 & 2 \\ 0 & \boxed{1} & 2 & 0 & 1 \\ 0 & 0 & 0 & \boxed{1} & 3 \end{array}\right] \xrightarrow{ins-0\ rows} \left[\begin{array}{cccc|c} \boxed{1} & 0 & 1 & 0 & 2 \\ 0 & \boxed{1} & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & 3 \end{array}\right]$.

   $\xrightarrow{change\ 0\ to\ -1} \left[\begin{array}{cccc|c} \boxed{1} & 0 & 1 & 0 & 2 \\ 0 & \boxed{1} & 2 & 0 & 1 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & 3 \end{array}\right]$. [**1**]

   $\mathrm{Sol}(A, b) = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 3 \end{bmatrix} + \alpha \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix} : \alpha \in \mathbb{F} \right\}$. [**1**]

2. Let $A, B \in \mathbb{F}^{m \times n}$ be such that $Ax = Bx$ for each $x \in \mathbb{F}^{n \times 1}$. Prove that $A = B$. [3]

   Solution: Suppose $Ax = Bx$ for each $x \in \mathbb{F}^{n \times 1}$. Then, $Ae_j = Be_j$ for $j = 1, 2, \ldots, n$. It implies that the jth column of A is equal to the jth column of B. So, $A = B$. [**3**]

3. Do there exist matrices $A, B \in \mathbb{C}^{n \times n}$ such that $AB - BA = (1 + 2i)I$? [2]

   Solution: $\mathrm{tr}(AB - BA) = \mathrm{tr}(AB) - \mathrm{tr}(BA) = 0$. [**1**]

   But $\mathrm{tr}((1 + 2i)I) = (1 + 2i)n$. Hence, there do not exist such matrices A and B. [**1**]

4. Let $A \in \mathbb{R}^{n\times n}$ be invertible and let $k \in \{2, \ldots, n-1\}$, where n is a natural number larger than 4. Construct a matrix $B \in \mathbb{R}^{k\times n}$ whose rows are any k rows chosen arbitararily from among the rows of A. Show that $\operatorname{rank}(B) = k$. [3]

   Solution: Since A is invertible, all its n rows are linearly independent. [**1**]

   Therefore, any k of these rows are also linearly independent. Then, in the RREF of B each row is a pivot so that $\operatorname{rank}(B) = k$. [**2**]

5. Let $A \in \mathbb{F}^{n\times n}$. If $\operatorname{rank}(A) = 1$, then prove the following:

   (a) There exist nonzero vectors $u, v \in \mathbb{F}^{n\times 1}$ such that $A = uv^*$. [2]

   Solution: Since $\operatorname{rank}(A) = 1$, there is only one linearly independent row in A. Then this row is a nonzero row. Take this row as v^* for $v \in \mathbb{F}^{n\times 1}$. All other rows are linear combinations of this row, i.,e., all other rows are scalar multiples of this row. Construct u by taking these scalars as respective components.
   Then, $uv^* = A$. [**2**]

   (b) $A^2 = \alpha A$ for some $\alpha \in \mathbb{F}$. [2]

   Solution: By Part (a), $A = uv^*$. Then, $A^2 = uv^*uv^* = u\,(v^*u)\,v^*$. As $v^*u \in \mathbb{F}$, we have $A^2 = (v^*u)\,uv^* = (v^*u)\,A$. The scalar $\alpha = v^*u$. [**2**]

6. Construct 3×3 matrices A and B such that $\det(A + B) \neq \det(A) + \det(B)$. [2]

   Solution: Take $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then, $A + B = I$.
   Now, $\det(A) + \det(B) = 0 + 0 = 0$ but $\det(A + B) = \det(I) = 1$. [**2**]
